

EDUCATION

Imperial College London (London, UK)

Sep 2025 – Sep 2026

M.Sc. Statistics

Thesis: Spectral Mixture Kernel Approximation with DeepRV for
Geospatial Bayesian Inference

Advisor: Oliver Ratmann

Pomona College (Claremont, California, USA)

Aug 2021 – May 2025

B.A. Mathematics and Spanish

Thesis: “Regression to the Mean”: The Confluence of Eugenics and Statistics in the 19th and 20th Centuries (distinction)

Advisors: Shahriar Shahriari, Jemma Lorenat, Susana Chávez-Silverman

Trinity College Dublin (Dublin, Ireland)

Jan 2024 – May 2024

Undergraduate studies in mathematics, Spanish, and Irish

RESEARCH EXPERIENCE

Predoctoral Scholar, Data and Democracy Lab

Sep 2026 – Present

University of Chicago (Chicago, Illinois, USA)

- Investigate the intersection of social choice theory and mathematical modelling
- Build and maintain simulations used to test for gerrymandering in electoral maps
- Analyze the legal landscape of gerrymandering and representation in the United States

Master's Thesis in Statistics

Jan 2026 – Sep 2026

Imperial College London (London, UK)

- Achieved distinction in the dissertation, reflecting a publishable contribution to research
- Introduced extension of approximate methods for Bayesian spatial statistics, allowing for full posterior sampling of spectral mixture kernels
- Applied novel methods to air quality data in Los Angeles
- Compiled review on neural surrogates for spatial inference with four colleagues
- Communicated findings in real-time to supervisors via written and oral presentations

Writer, Copyeditor, & Survey Data Analyst

Sep 2025 – Jun 2026

Felix, Imperial College London Newspaper (London, UK)

- Contributed regular articles including opinions/editorials and book reviews
- Prepared weekly newspapers via timely proofreading, licensing, and layout management
- Critiqued and edited creative writing submissions to literary magazine Phoenix
- Managed and visualized confidential data from student body surveys

Humanities Studio Fellow

Jun 2024 – May 2025

Pomona College (Claremont, California, USA)

- Selected as one of six fellows to contribute to interdisciplinary conversations on misinformation, immigration and diaspora, digital privacy, and gender inequality
- Initiated discussions on the limits of AI and LLMs in universities
- Adapted reports on statistics and machine learning to non-technical audiences

Undergraduate Thesis in Mathematics

Jun 2024 – May 2025

Pomona College (Claremont, California, USA)

- Investigated the intersection of statistical methods and eugenics in Britain circa 1900
- Built database of 250 statistics textbooks, ensuring compliance with publishing licenses
- Evaluated presence of eugenic ideology in statistics via text mining/sentiment analysis

Undergraduate Researcher

May 2023 – Jul 2023

Carnegie Mellon University (Pittsburgh, Pennsylvania, USA)

- Coordinated team of four colleagues researching zero-forcing processes in graph theory
- Established propagation time bounds for new method of zero forcing on common graphs
- Devised algorithm in Python/SAGE to model long-term behavior of zero-forcing

**TEACHING
EXPERIENCE****Pomona College Department of Mathematics and Statistics**

as course mentor and grader, responsible for 30-80 students per semester

MATH158: Statistical Linear Models

Spring 2025

MATH113: Number Theory and Cryptography

Fall 2024

MATH031: Calculus II

Fall 2023

MATH103: Combinatorial Mathematics

Spring 2023

MATH060: Linear Algebra

Fall 2022

1-2-1 Math Program

as teaching assistant, responsible for 15 students in groups of 5

Tangelo, an exploratory math curriculum for high school students

Summer 2022

PUBLICATIONS

2. Elias, Olivia, Ian Farish, **Emrys King**, Josh Kyei, and Ryan Moruzzi Jr. “Leaky Positive Semidefinite Forcing on Graphs.” *Involve, a Journal of Mathematics* 18, no. 4 (2025): 719–35.
<https://doi.org/10.2140/involve.2025.18.719>.
1. **King, Emrys**. “Regression to the Mean?: The Confluence of Eugenics and Statistics in the 19th and 20th Centuries.” Pomona College, 2025.
https://scholarship.claremont.edu/pomona_theses/347.

PRESENTATIONS

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| “DeepRV with Spectral Mixture Kernels: A Novel Neural Surrogate Method for Geospatial Bayesian Inference”
Imperial College London Department of Mathematics, MSc Statistics Research Project Presentations; London, UK. 20-minute oral presentation. | 8 Sep 2026 |
| “SM-DeepRV: A Neural Surrogate for Geospatial Gaussian Processes”
Imperial College London Department of Mathematics, MSc Statistics Poster Presentations; London, UK. Poster presentation. | 7 July 2026 |
| “The Historical Confluence of Statistical Linear Modelling and Eugenics”
Pomona College Department of Mathematics and Statistics Senior Seminar; Claremont, CA, USA. 20-minute oral presentation. | 7 Mar 2025 |
| “The Mixing of Eugenics and Statistics in English-Language Pedagogy in the 20th Century.”
Claremont Seminar in the History and Philosophy of Mathematics, The Alchemy of Mixing Mathematics Workshop; Claremont, CA, USA. 20-minute oral presentation. | 15 Feb 2025 |
| “Positive Semidefinite Leaky Zero Forcing: Preliminary Report.”
Joint Mathematics Meetings, San Francisco, CA. Poster presentation. | 4 Jan 2024 |

- “Efficacy of Community Learning Models in STEM Classrooms”
Pomona College IDEAL (Inclusion, Diversity, Equity, And Learning)
Symposium; Claremont, CA, USA. Poster presentation. 6 Dec 2023
- “Positive Semidefinite Leaky Forcing – A New Variation of Zero Forcing”
SACNAS National Diversity in STEM Conference and the Modern
Mathematics Workshop; Portland, OR, USA. Poster presentation. 27 Oct 2023
Awarded for excellence in undergraduate poster. Travel funded by
Institute for Mathematical and Statistical Innovation (IMSI).

SERVICE & VOLUNTEERING	Hampshire College Transfer Application Reader remote, following the announcement of Hampshire College’s closure	Apr 2026
	Learning Analytics Hackathon Mentor Imperial College London (London, England, UK)	Nov 2025
	Department of Mathematics & Statistics Student Liaison Pomona College (Claremont, California, USA)	Sep 2023 – May 2025
	Incoming Student Orientation Team Member Pomona College (Claremont, California, USA)	Aug 2022
	Legacy Youth Leadership Program Docent Birmingham Civil Rights Institute (Birmingham, Alabama, USA)	Feb 2020 – May 2021
WORKSHOPS & COURSES	MSc Statistics Data Science Challenge Imperial College London (London, England, UK)	Jul 2026
	Big Data for Social Good Harvard Online	Jun – Jul 2026
	DSSI Data Science for Social Good Datathon Harvey Mudd College (Claremont, California, USA)	Apr 2025
	Machine Learning Workshop Series Harvey Mudd College (Claremont, California, USA)	Oct – Nov 2024
HONORS	Bruce Jay Levy Senior Prize in Mathematics	2025
	Sigma Xi	2025
	NSF Graduate Research Fellowship Program – Honorable Mention	2025
	Pomona College Scholar	2023, 2025
	Llewellyn Bixby Prize in Mathematics	2023
	Sigma Delta Pi	2023
	NewGen Peacebuilder	2021